

INTERMEDIATE RESEARCH PRESENTATION

District-Level Dengue Outbreak Early Warning System for Sri Lanka

BSc (Hons) in Software Engineering — Faculty of Science, University of Kelaniya

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Background & Problem Statement

Dengue is a recurring public-health challenge in Sri Lanka - incidence varies sharply across time, geography, and climatic zones, yet surveillance is largely retrospective, describing infections only after they occur. Effective prevention needs a mechanism that converts historical case and climate data into reliable, location-specific warnings before a surge becomes severe.

1

Limited geographical generalisation

Most studies focus on one or a few districts; findings rarely generalise nationally.

2

Delayed climate effects under-used

Climate–dengue relationships are often lagged, not same-week, and rarely studied locally.

3

No fair comparative benchmark

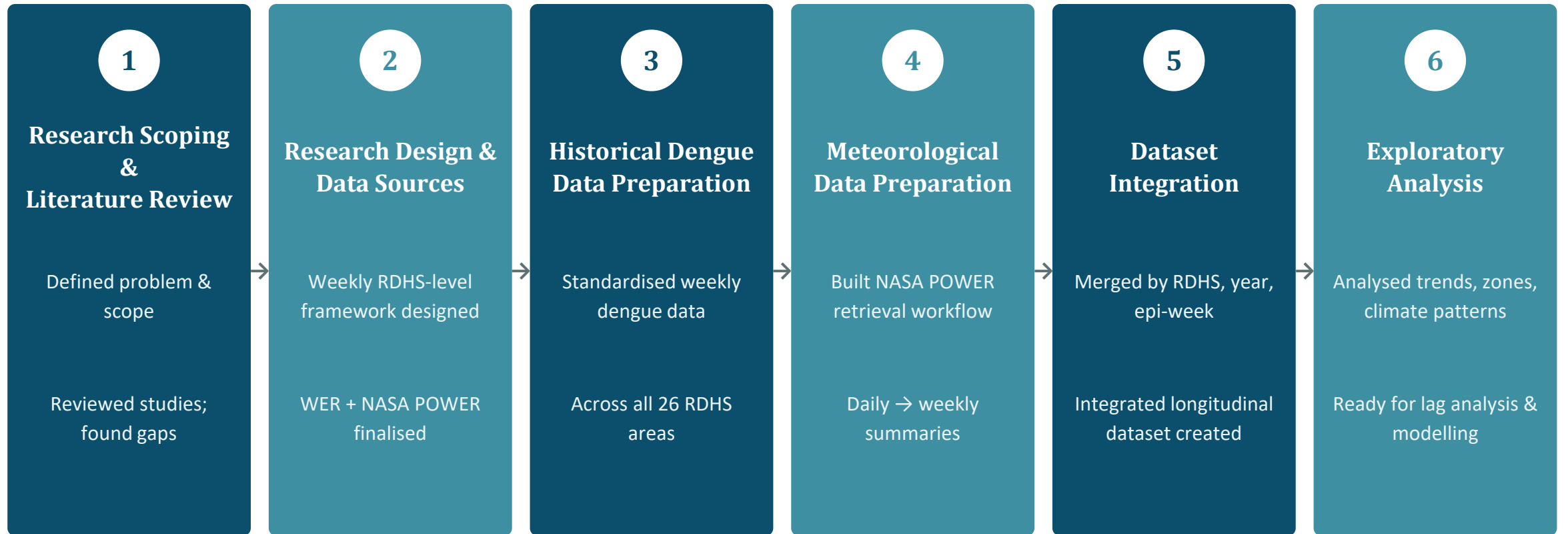
Statistical / ML / deep-learning comparisons rarely share one dataset, horizon, and validation.

4

Missing decision-support layer

Accurate forecasts alone aren't enough — outputs must become actionable alerts.

Research Progress to Date



Where We Are Today

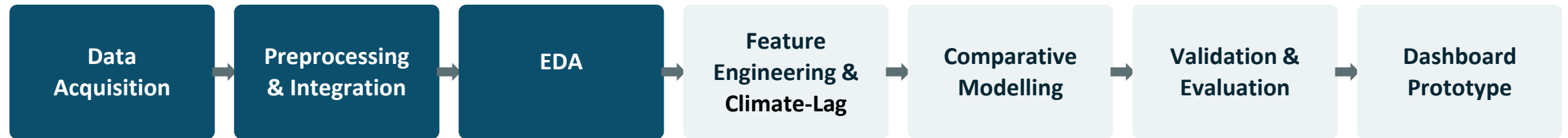
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methodology stages
completed

26

RDHS areas in the
integrated dataset

Methodology Progress



On track: the integrated dataset and exploratory analysis are complete. Comparative modelling, outbreak classification, and the dashboard are the focus of the remaining research.

Theoretical Foundations

Dengue transmission is both temporally dynamic and spatially heterogeneous — shaped by historical case patterns, seasonal conditions, climate, and mosquito-breeding biology. Sri Lankan evidence establishes three foundations for this research:

1

A persistent national problem

Fernando et al. (2024) trace dengue's epidemiological burden in Sri Lanka, 2000–2020, including major peaks in 2017 & 2019 — confirming the need for continuous, proactive surveillance, not one-off response.

2

Behaviour varies spatially & seasonally

Prabodanie et al. (2020) found wet-zone districts often show two epidemic waves/year vs. one in dry-zone coastal districts; Edirisinghe (2017) & Sun et al. (2017) confirm spatial hotspot/cold-spot clustering.

3

Climate effects are delayed, not immediate

Withanage et al. (2018) and Uduwanage et al. (2025) show weak same-period weather links that strengthen once precipitation, temperature & humidity are time-shifted — the theoretical basis for climate-lag analysis here.

Existing Approaches & Related Work

Research	What the Research Shows	Limitation
Prabodanie et al. (2020)	Weekly dengue patterns differ across districts and climatic regions in Sri Lanka	Descriptive analysis only; no forecasting model
Withanage et al. (2018)	Lagged rainfall, temperature, and humidity can support dengue forecasting	Gampaha only; monthly data
Uduwanage et al. (2025)	Compared CNN, XGBoost, and SARIMAX; lagged climate variables improved prediction	Selected districts; data aggregated monthly
Rahman et al. (2025)	LightGBM with SHAP was used for interpretable dengue early warning	Conducted in Bangladesh; requires Sri Lankan validation
Chen & Moraga (2026)	LSTM was used for weekly dengue forecasting using climate and mobility data	Conducted in Brazil; mobility data unavailable in our study

Research Gap → This Study

1

Limited geographic coverage

Most studies cover only a few districts

2

Monthly, not weekly, granularity

Less operational lead-time for early warning

3

Weather-data availability

Prior work constrained to selected districts / stations

4

Limited systematic climate-lag analysis

Optimal lags per area/variable remain unclear

5

No common model comparison

SARIMAX, ML, DL rarely evaluated on one dataset

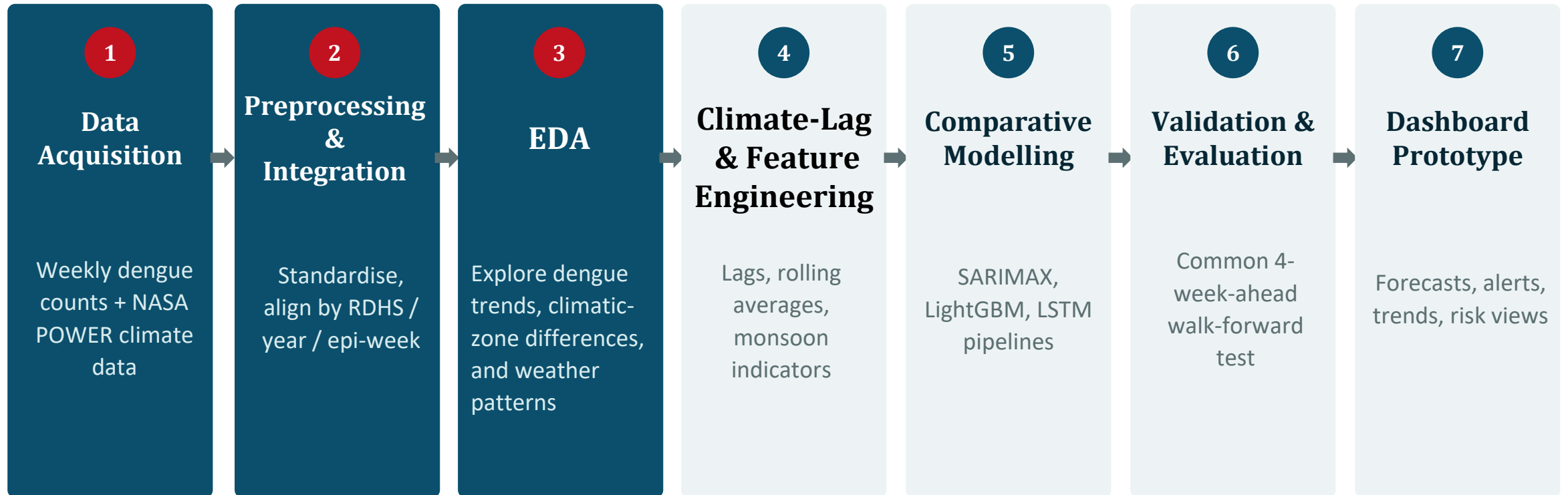
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Limited operational decision support

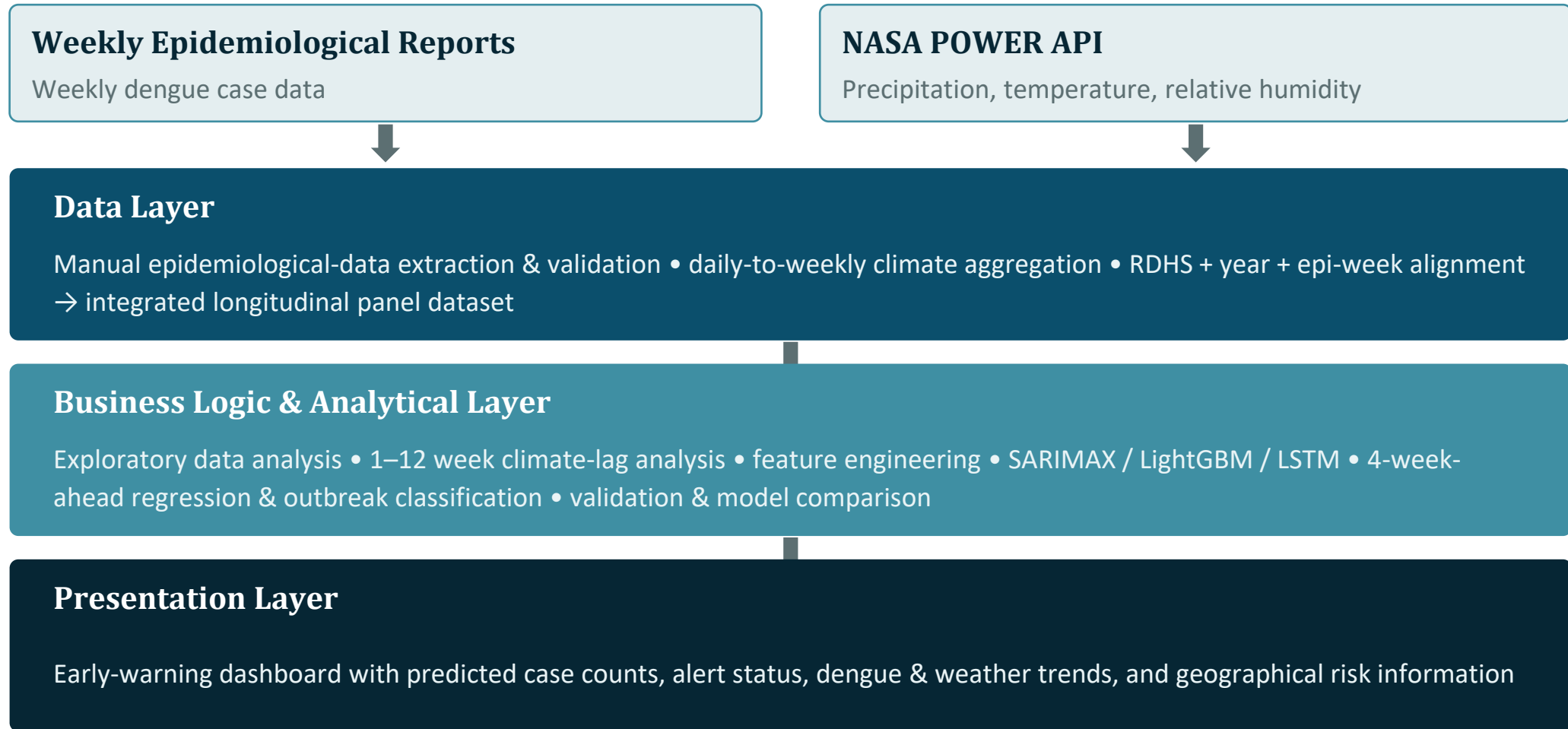
Forecasts rarely reach an actionable dashboard

Methodology Overview

A quantitative, experimental research design combined with software-prototype development — seven stages sharing one common forecast horizon and evaluation procedure.



System Architecture



Validation Strategy & Evaluation Metrics

Time-Aware Walk-Forward Validation

Random splitting mixes past & future — not valid for forecasting. Instead, the training window expands forward and is tested on the next unseen year:

Train	Test
2016–2020	2021
2016–2021	2022
2016–2022	2023
2016–2023	2024

Regression Metrics

MAE: Average absolute error

RMSE: Penalises larger errors more

MAPE: Error as % of observed value

Bias: Systematic over/under-prediction

Classification Metrics

Precision: Share of alerts that were real outbreaks

Recall: Share of real outbreaks correctly flagged

F1-score: Balance of precision & recall

ROC-AUC: Ability to separate alert vs. non-alert

Data Sources & Coverage

Data Sources

Epidemiological

Ministry of Health Epidemiology Unit — Weekly Epidemiological Reports, indexed by year, epi-week, and RDHS area.

Meteorological

NASA POWER API — precipitation, temperature, relative humidity via representative RDHS-area coordinates, aggregated daily → weekly.

Coverage Snapshot

26

RDHS - Regional Director of Health Services

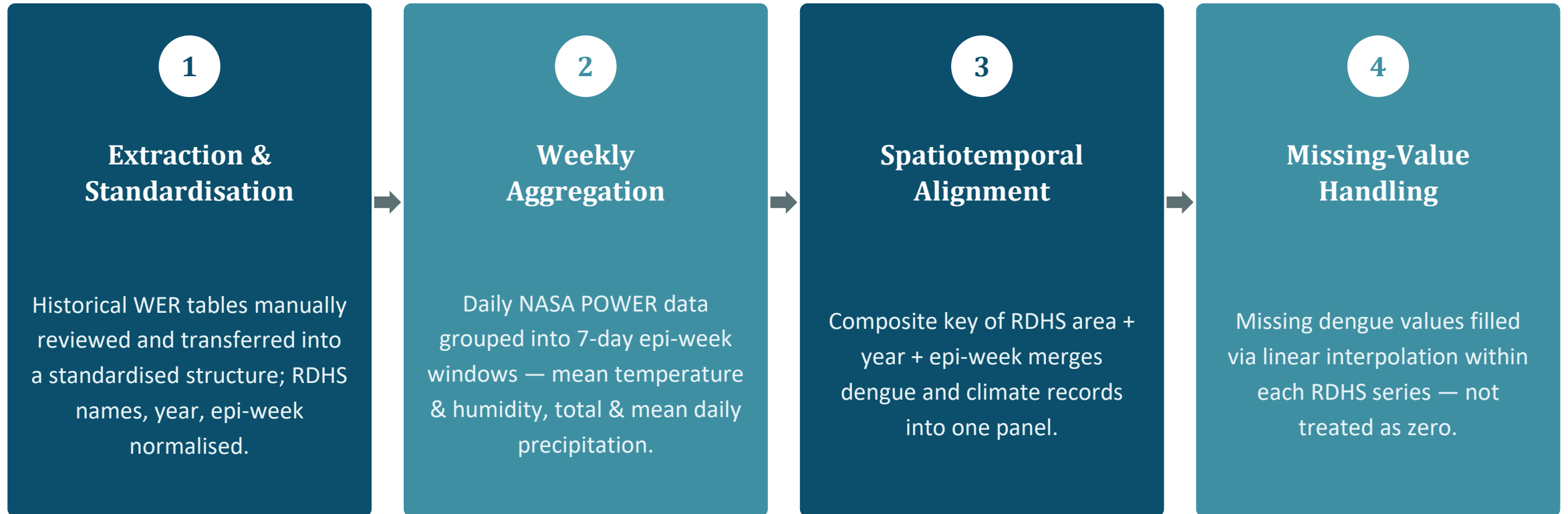
2016–2026

Weekly observations, Jan 2016 – 31 May 2026

3

Climatic zones represented: Wet, Dry, Intermediate

Preprocessing & Integration Pipeline



Dataset Access

XLSX

Final Integrated RDHS-Week Dataset

Final_weather_and_dengue_cases_data_set.xlsx

National Dengue Case Trend

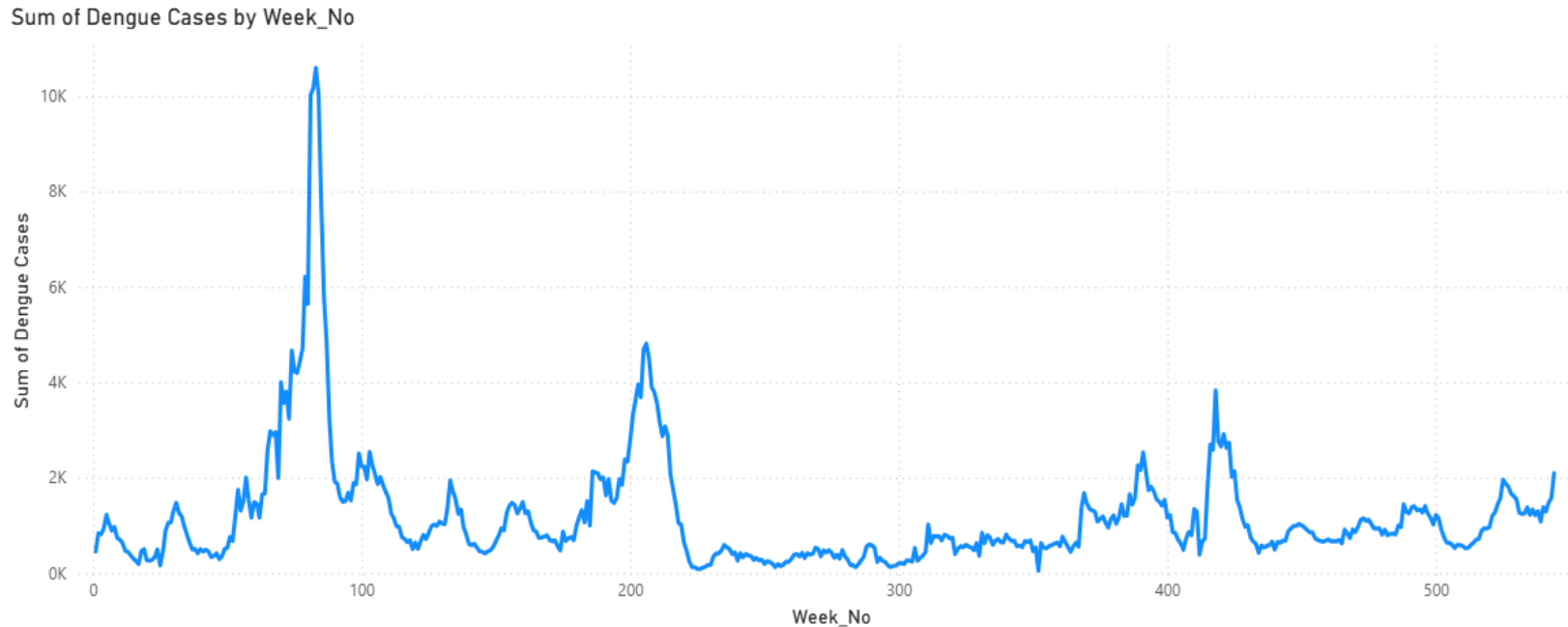


Figure 4.1 — Aggregated weekly dengue cases across the full study timeline (2016–2026)

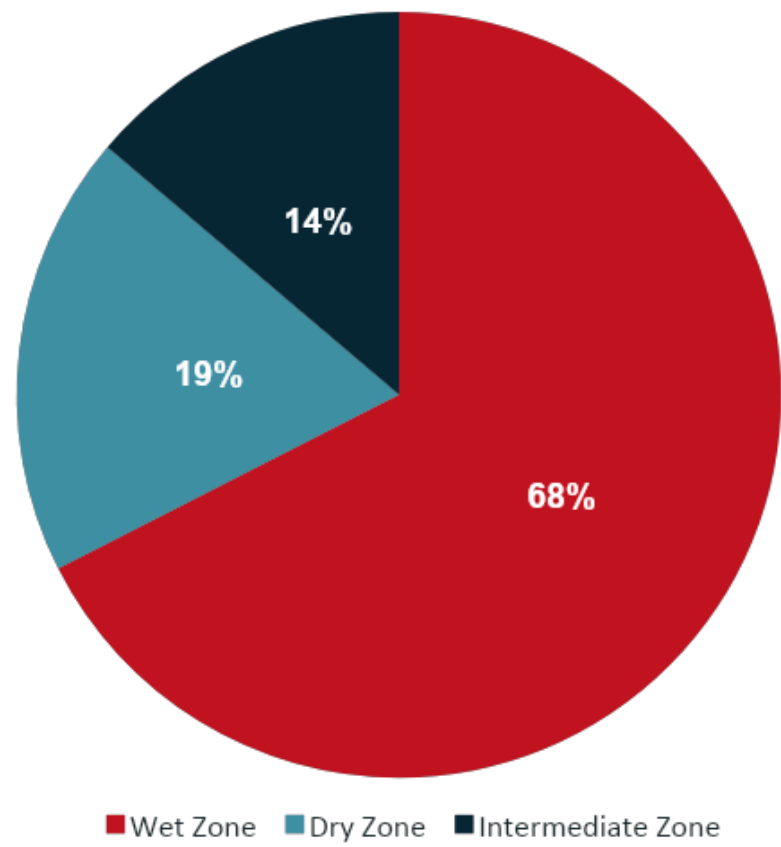
~10,500

peak weekly cases - the largest surge in the series

Highly irregular, non-uniform incidence - sharp epidemic peaks separated by long, low-incidence periods.

The prediction task must capture both longer-term temporal structure and sudden epidemic peaks - a purely static approach is unlikely to work.

Dengue Burden by Climatic Zone



Climatic Zone	Approx. Cases	Share
Wet Zone	437,000	67.5%
Dry Zone	121,000	18.69%
Intermediate Zone	89,000	13.8%

The Wet Zone alone accounts for roughly two-thirds of cumulative dengue cases in the study dataset - descriptive motivation for retaining climatic-zone information in later feature engineering, though this alone does not establish causation.

Climate Variables vs. Dengue Cases

National Weekly Average Precipitation and Dengue Cases

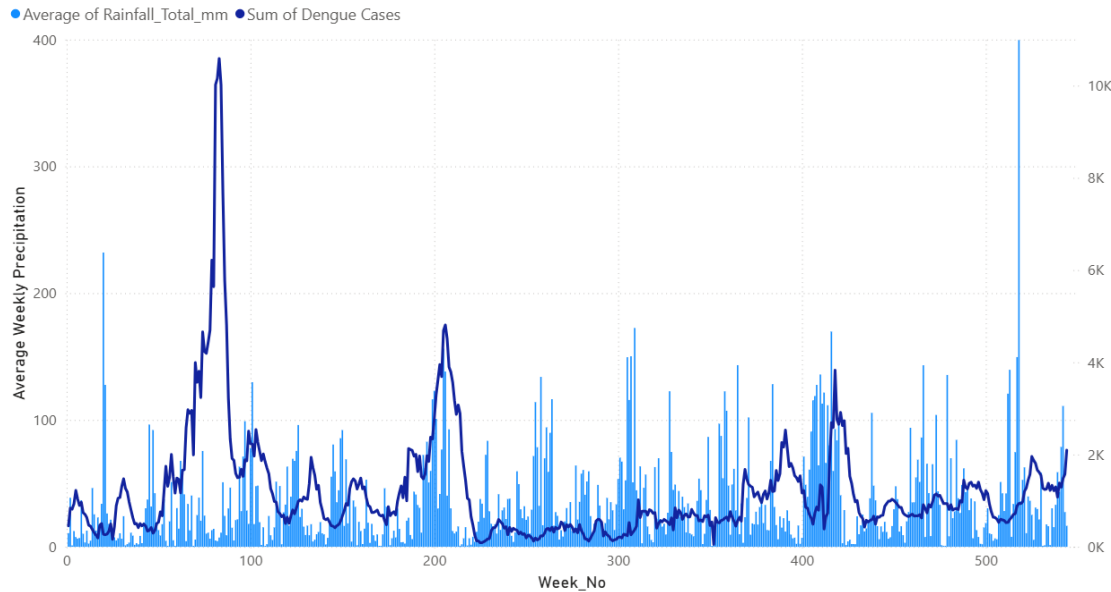


Figure 4.3 — Precipitation vs. dengue

National Weekly Average Temperature and Dengue Cases

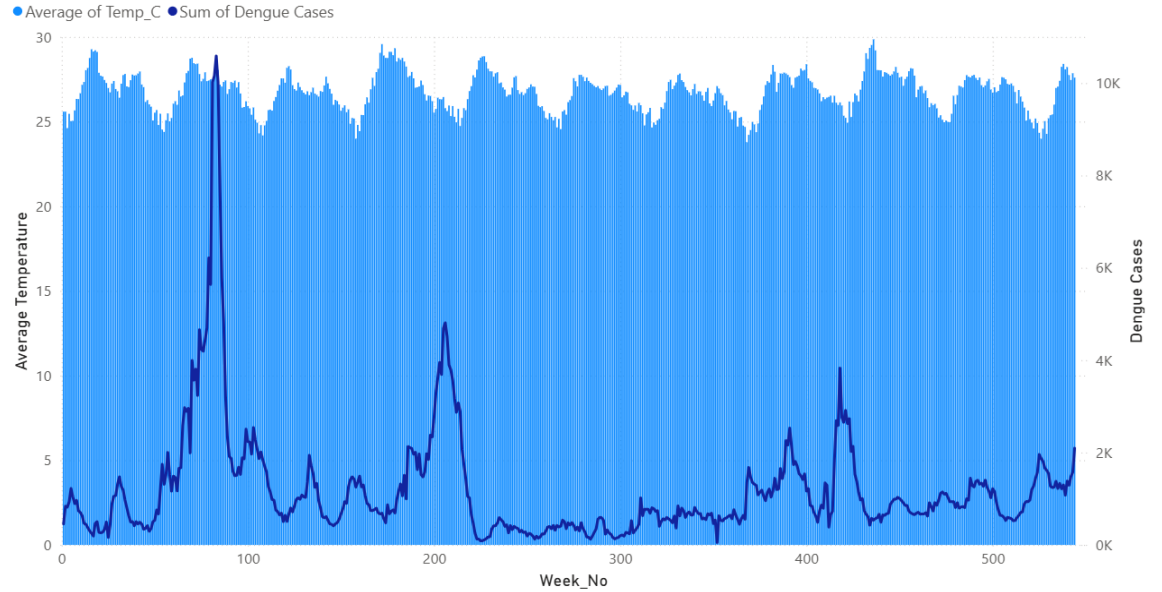


Figure 4.4 — Temperature vs. dengue

No simple same-week relationship: precipitation shows substantial week-to-week variation, while temperature and relative humidity vary within narrower ranges. Major dengue peaks do not consistently coincide with same-week climate peaks. National averaging across 26 RDHS areas can mask local and delayed responses - motivating the planned RDHS-level, 1–12 week climate-lag analysis.

Implementation Status

Historical WER dengue-data preparation	RDHS-specific 1–12 week climate-lag analysis
Completed	To be completed
NASA POWER meteorological-data retrieval workflow	Comparative SARIMAX / LightGBM / LSTM evaluation
Completed	To be completed
Daily-to-weekly meteorological aggregation	Four-week-ahead outbreak-classification evaluation
Completed	To be completed
RDHS-week epidemiological & climatic data integration	Complete dashboard integration
Completed	To be completed
Exploratory data analysis	Formal dashboard usability evaluation
Completed	To be completed

What We've Built So Far

Functional Verification

- ✓ Weekly dengue observations structured consistently across all 26 RDHS areas
- ✓ NASA POWER retrieval utility obtains daily precipitation, temperature & humidity
- ✓ Daily meteorological data converts cleanly into weekly summaries
- ✓ Dengue & climate records align correctly by RDHS area, year & epi-week
- ✓ Integrated dataset supports exploratory visualisation & feature engineering

Tools & Artifacts Completed

Streamlit weather-retrieval utility

Automates NASA POWER API calls via Python requests

Integrated RDHS-week dataset

26 areas × weekly panel, 2016–2026, ready for modelling

Exploratory analysis notebooks

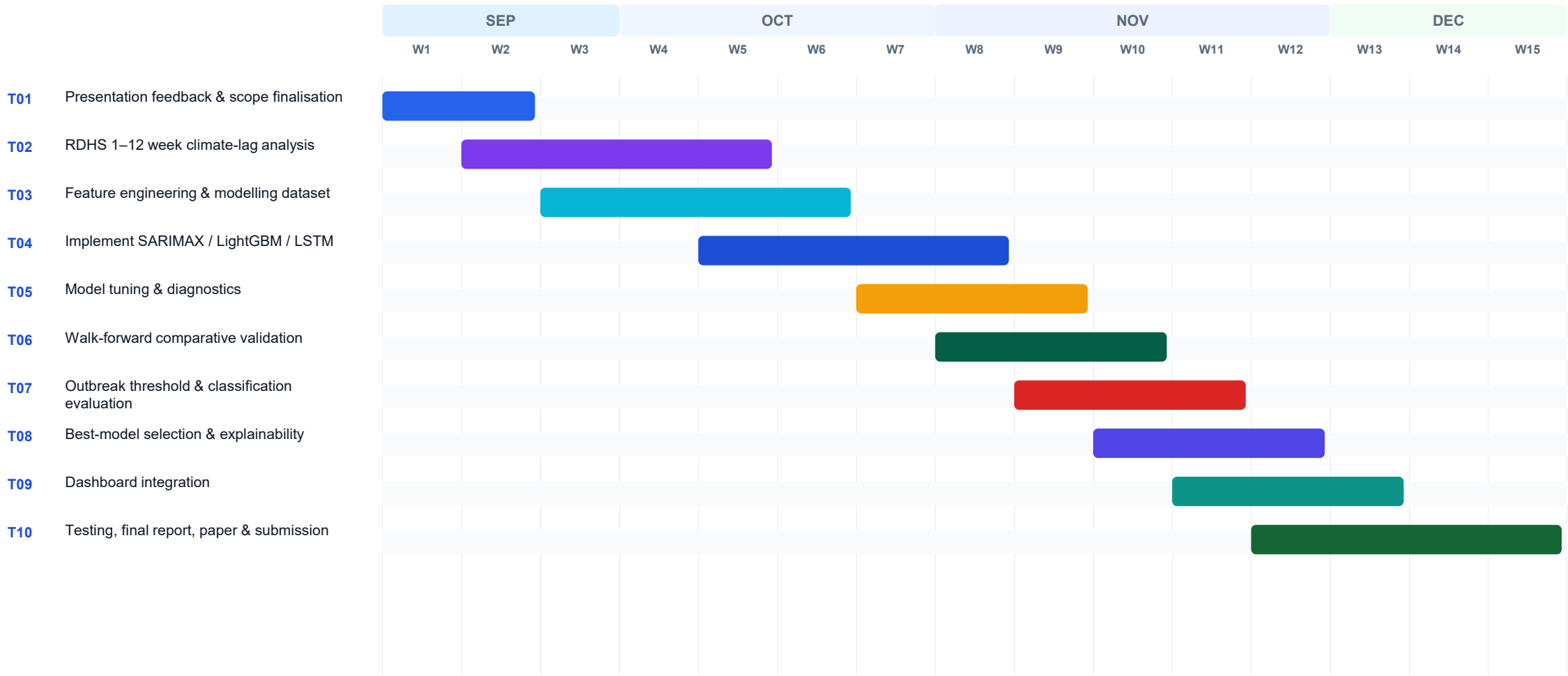
Trend, zone, and climate-relationship visualisations

GitHub repository

Source code, documentation, and experiment version history

Revised Semester 2 Timeline

14 September – 25 December 2026 | 15-week completion plan



Semester 2 completion

- 1 Complete RDHS-level 1–12 week climate-lag analysis.
- 2 Finalise lag, rolling, monsoon and climate-zone features.
- 3 Train, tune and compare SARIMAX, LightGBM and LSTM.
- 4 Evaluate four-week case forecasts and outbreak-alert classification.
- 5 Integrate the selected output into the dashboard and complete final evaluation.

CONCLUSION

Interim Milestone Summary

- 1 The project has moved from concept to a working data foundation for dengue early warning in Sri Lanka.
- 2 A 2016–2026 RDHS-week dataset covering all 26 reporting areas has been integrated and internally verified for alignment.
- 3 Exploratory analysis confirms strong temporal variation, climatic-zone differences, and no simple same-week climate signal.
- 4 Research gap addressed: a weekly RDHS-level framework combining NASA POWER weather data, 1–12 week lag analysis, SARIMAX/LightGBM/LSTM comparison, outbreak classification and dashboard support.
- 5 Novelty: one Sri Lankan framework with dual outputs — four-week case-count forecasting and localized outbreak-alert classification.
- 6 Semester 2 will complete lag analysis, feature engineering, modelling, validation, dashboard integration and final evaluation.

QUESTIONS & DISCUSSION

Q&A

District-Level Dengue Outbreak Early Warning System for Sri Lanka
